

WEB SYSTEMS AND TECHNOLOGIES

FINAL PROJECT/EXAM

Project Overview

This final project is designed to strengthen your understanding of modern web development by recreating a functional module. Your task is to identify and recreate ONE CRUD (Create, Read, Update, Delete) module from an existing capstone project or a topic that you like as long as it will utilize CRUD.

You may use the same database schema and topic from your Information Management 1 (final project by sir RR)

1. CRUD Functionality

- Create: Ability to add new records with form validation
- Read: Display records in a table or list with search/filter capabilities
- Update: Edit existing records with confirmation
- Delete: Remove records with safety measures (confirmation dialog)

If the original project used outdated technologies, recreate it with modern alternatives. (optional)

Technical Documentation (follow this format)

1. Project Overview

- Original Project Title: [Name of the Capstone project you selected]
- Original Authors/Source: [Link to published paper or repository]
- Module Scope: A brief description of what this specific module does.
- Rationale: Why did you choose this specific module? What technical challenge does it solve? How does it differ from the original implementation?

2. Technology Stack

- Frontend Framework: (e.g., React 18, Vue 3, Alpine.js)
- Styling: (e.g., Bootstrap 5, Tailwind CSS, CSS Modules)
- Backend Framework: (e.g., Laravel 11, Node.js/Express, Django)
- Database: (e.g., MySQL 8.0, PostgreSQL, MongoDB)
- Additional Libraries: (e.g., Axios, Prisma, SQLAlchemy)

3. Database Design

Include an Entity-Relationship Diagram (ERD)

- Only the tables/collections relevant to this module
- Primary and foreign keys
- Relationships (one-to-many, many-to-many, etc.)
- Data types for each field

4. Module Features

Describe each CRUD operation with examples:

Create

- Form validation rules

- Error handling

Read ... Update... Delete...

5. Comparison with Original Implementation (optional)

Include a table or narrative comparing:

- Technology used in original vs. your implementation
- Key improvements made

6. Individual Reflection (Mandatory for All Members)

- What was the most difficult part of interpreting the original author logic?
- What technical challenges did you face during recreation?
- How would you improve this module further?

Submission Requirements

1. Create a public GitHub repository with all source code (src, database, readme file, documentation)
2. Include complete database setup files.
3. Provide detailed README.md with installation instructions.